Presentation title goes here, stated as the main claim

Subtitle: one sentence on what the audience will know by the end of the talk, kept to two lines at most

Presenter Name, Lab or Group, University of Southern California
Month DD, YYYY

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USC Viterbi School of Engineering Thomas Lord Department of Computer Science

Contents

01
Background
The problem, why it matters, and what was known

02
Approach
The core idea and how it differs from prior work

03
Setup
Data, baselines, metrics, and what counts as success

04
Results
The main measurements, with uncertainty and controls

05
Analysis
Where it works, where it breaks, and why it does

06
Next steps
Open questions, the timeline, and what you need

Setup table: the pieces of the system

Component	Configuration
Input	Dataset name, size, version and where it came from
Method	Algorithm or tool, version, and key settings used
Baseline	What you compare against, run under equal setup
Hardware	Machine, CPU or GPU, memory, OS and node type
Metric	Primary measure and how it is computed
Repeats	Number of runs, seeds, and how runs are ordered or paired

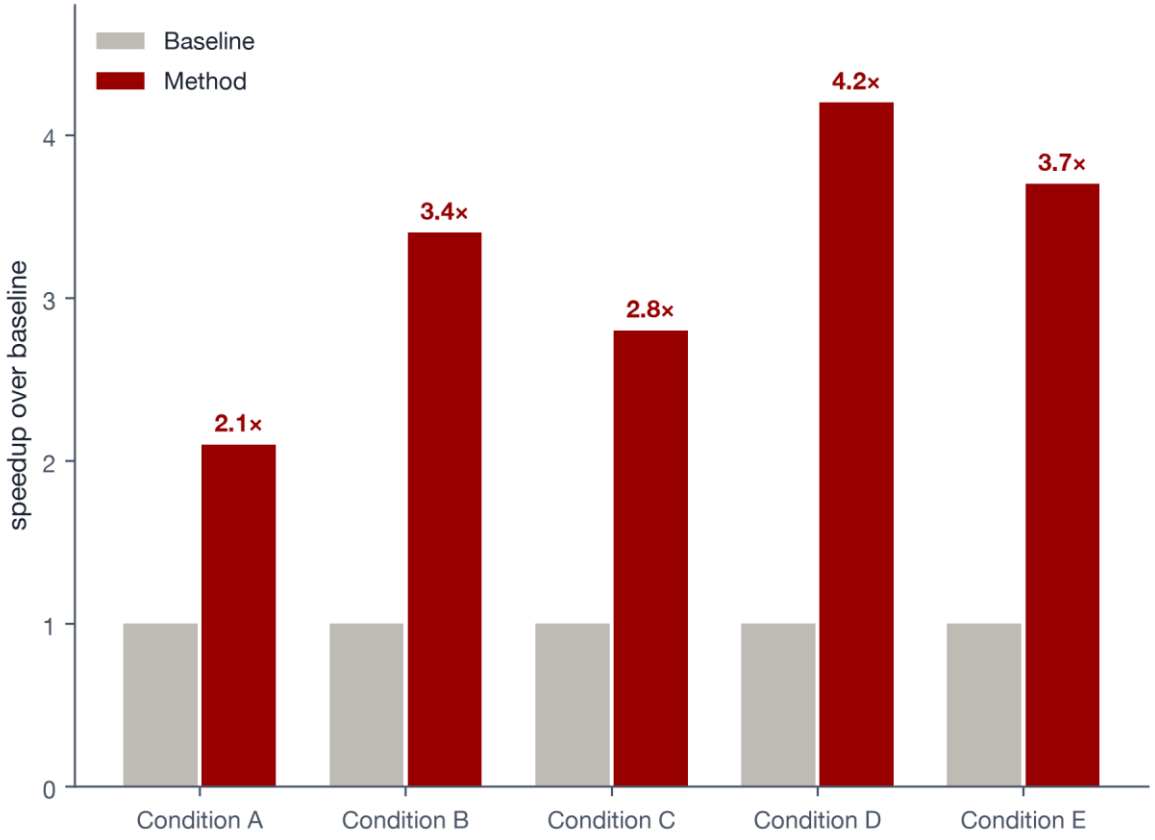
Footnote: one sentence that rules out the most likely misreading of this setup table.

Headline: the method beats the baseline everywhere

#	Condition	Value
A	2.1	±0.1
B	3.4	±0.2
C	2.8	±0.2
D	4.2	±0.3

Read the graph left to right: one pair of bars per condition, baseline in gray, method in cardinal.

Takeaway: the one sentence the audience should repeat.

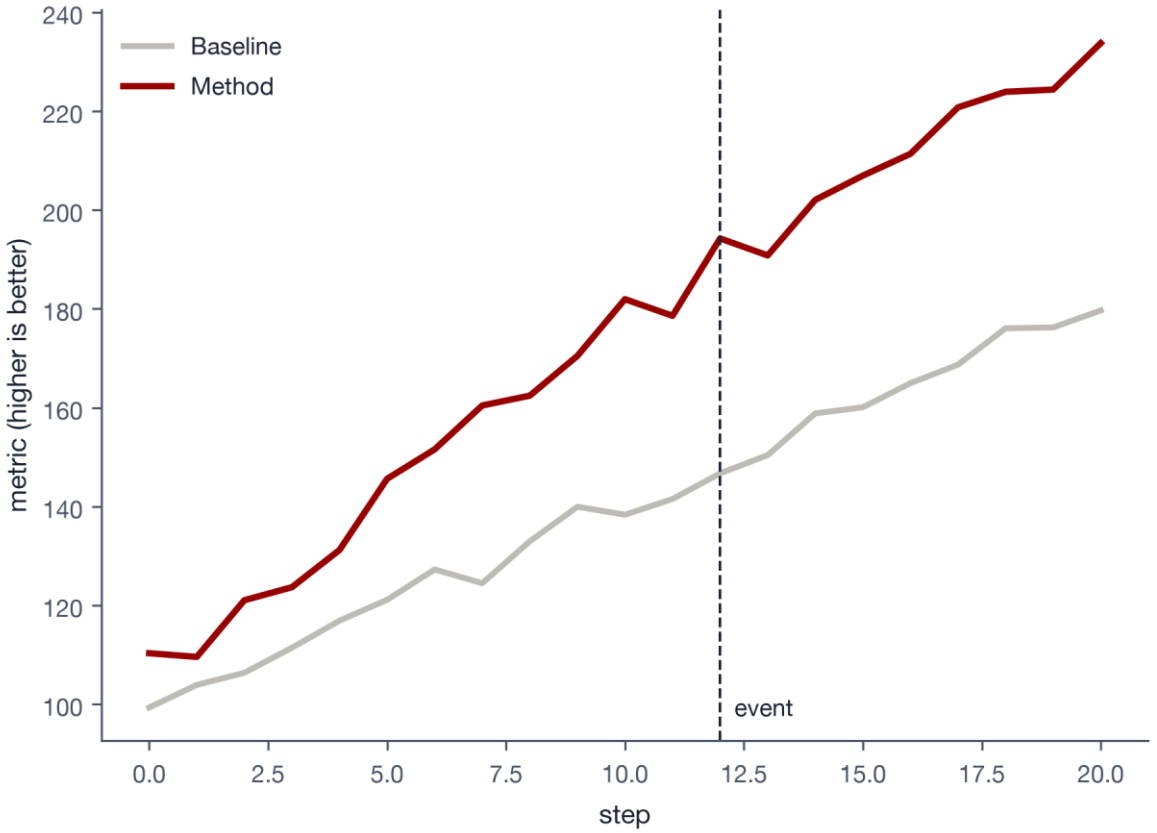


Trend: the gap opens over time, then holds

#	Condition	Value
A	2.1	±0.1
B	3.4	±0.2
C	2.8	±0.2
D	4.2	±0.3

Read the graph left to right: the metric over steps; the dashed line marks the event that matters.

Takeaway: what changed after the event, in one line.

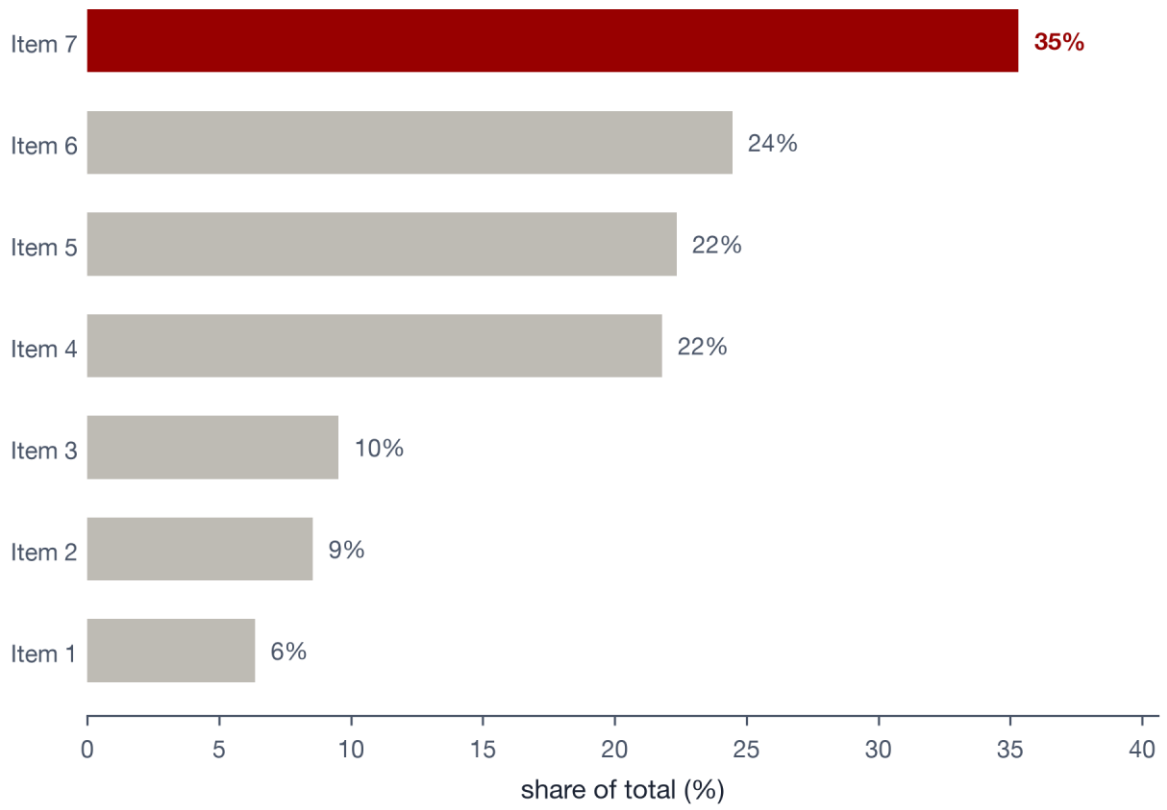


Ranking: one item dominates the total

#	Condition	Value
A	2.1	±0.1
B	3.4	±0.2
C	2.8	±0.2
D	4.2	±0.3

Read the bars top to bottom: sorted by share, the largest highlighted in cardinal, the rest gray.

Takeaway: why the top item matters most for the method.



Comparison table with two readings below

Metric vs outcome (correlation)	Metric A	Metric B	Metric C	Avg
Condition 1 (description)	0.82	0.41	0.10	0.44
Condition 2 (description)	0.75	0.38	0.05	0.39
Overall (all runs)	0.79	0.40	0.08	0.42

First reading of the table

Two sentences explaining what the first pattern in the table means and which row or column shows it.

Second reading of the table

Two sentences on the second pattern, ideally one that surprises or limits the first reading.

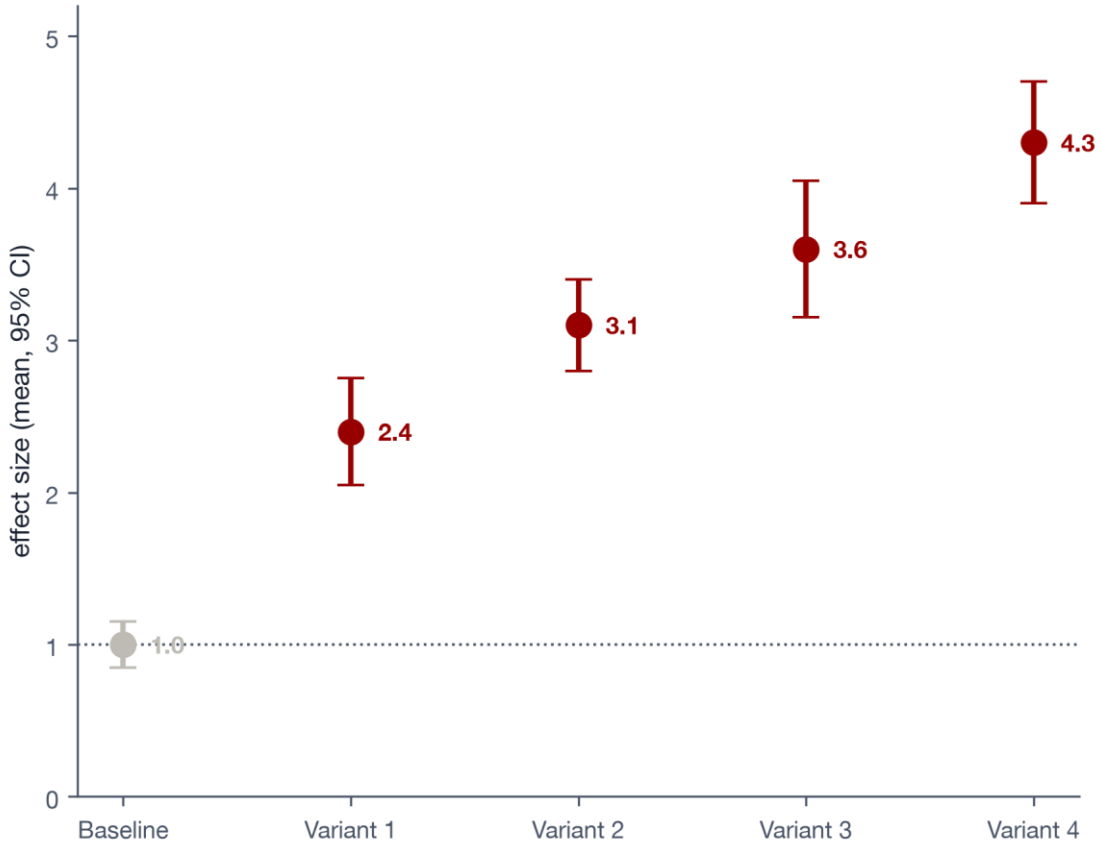
Key count: how many runs met the bar

4 of 5

conditions passed the pre-registered bar (state it)

Run (set)	Value	Interval
Run 1 (A)	2.4	±0.35
Run 2 (B)	3.1	±0.30
Run 3 (C)	3.6	±0.45
Run 4 (D, ref)	4.3 ×	±0.40 CI

One line saying what the count does and does not show.



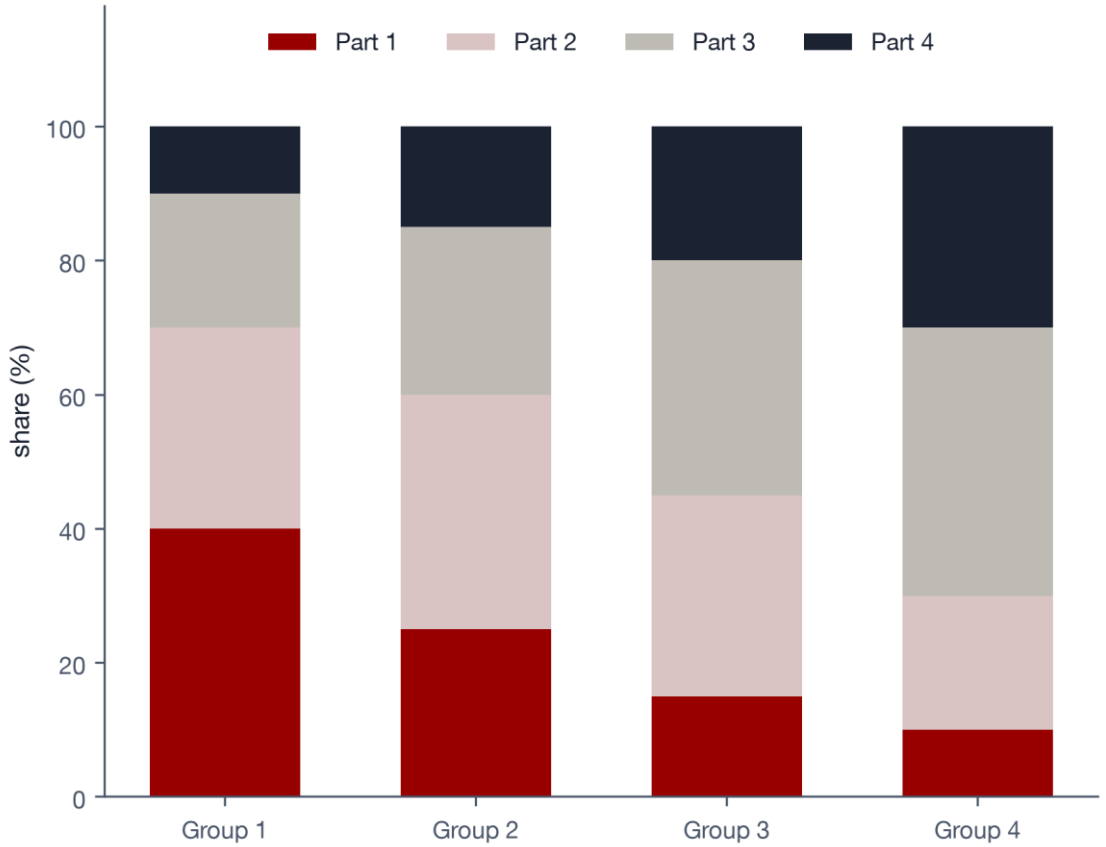
Key count: how many runs met the bar

4 of 5

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Run 4 (D, ref)	4.3 ×	±0.40 CI

One line saying what the count does and does not show.



Same input. Two different outcomes.

CASE A · BASELINE

1.00

label for the number above

→ **120**

outcome for case A

CASE B · METHOD

4.24

label for the number above

→ **28**

outcome for case B

Same input. Two different outcomes.

CASE A · BASELINE

1.00

label for the number above

→ 120

outcome for case A

CASE B · METHOD

4.24

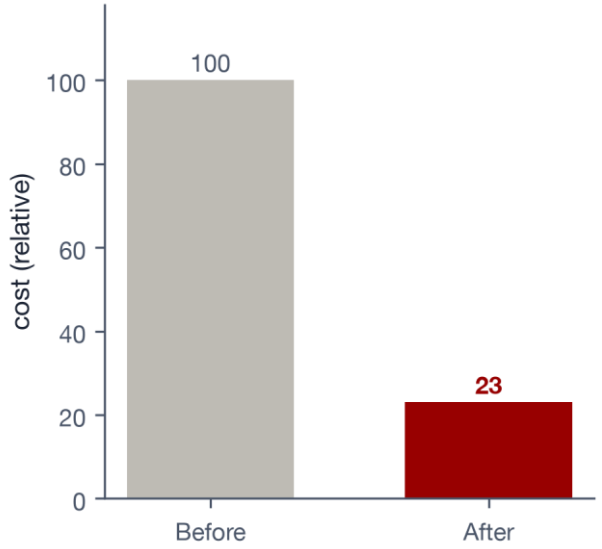
label for the number above

→ 28

outcome for case B

Side-by-side check: where the two cases differ

At checkpoint	Case A	Case B	Differs?
Property one	0.61	0.60	No
Property two	0.44	0.43	No
Property three	1.20	1.18	No
Property four	0.87	0.85	No
Property five	12.0	3.1	Yes
Property six	0.10	0.11	No
Property seven	5.2	5.0	No



Caption: what the small graph adds to the table, in one line.

The result in three numbers

4.24× vs 1.00×

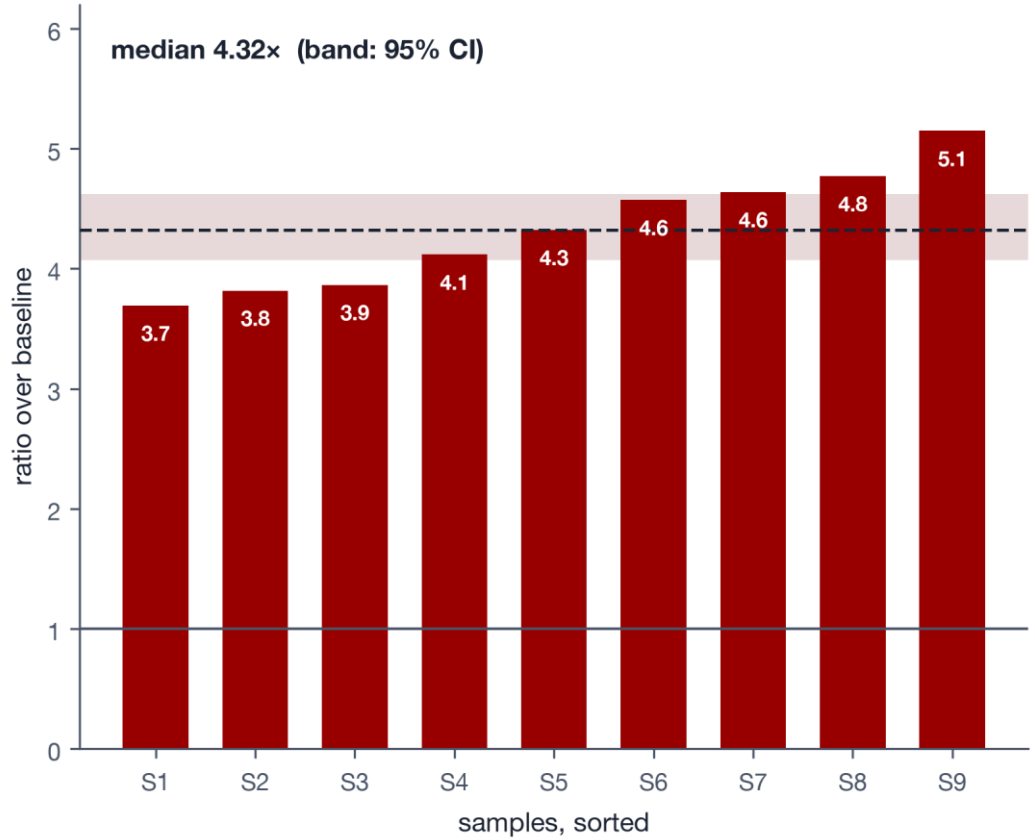
Headline ratio: method over baseline

0 of 715k

Second number: the check that must hold for the first to count

1.3× once

Third number: the cost or caveat



The result in three numbers

4.24× vs 1.00×

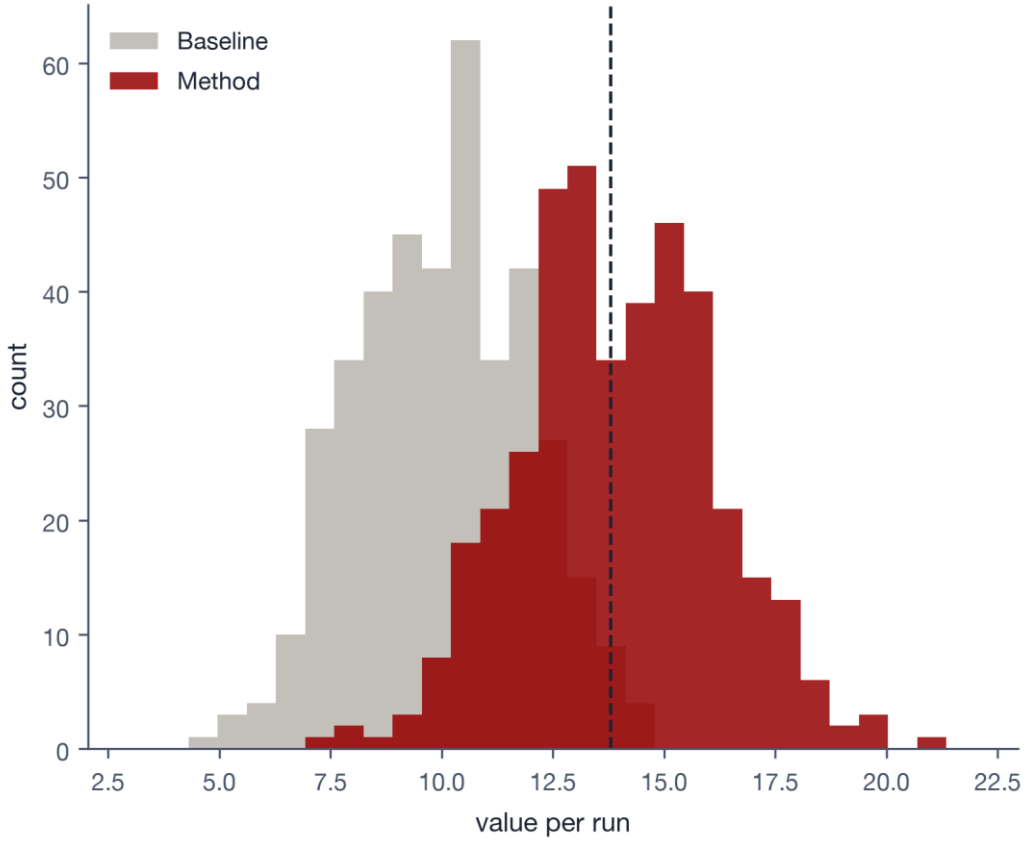
Headline ratio: method over baseline

0 of 715k

Second number: the check that must hold for the first to count

1.3× once

Third number: the cost or caveat



Elimination table: what the cause is not

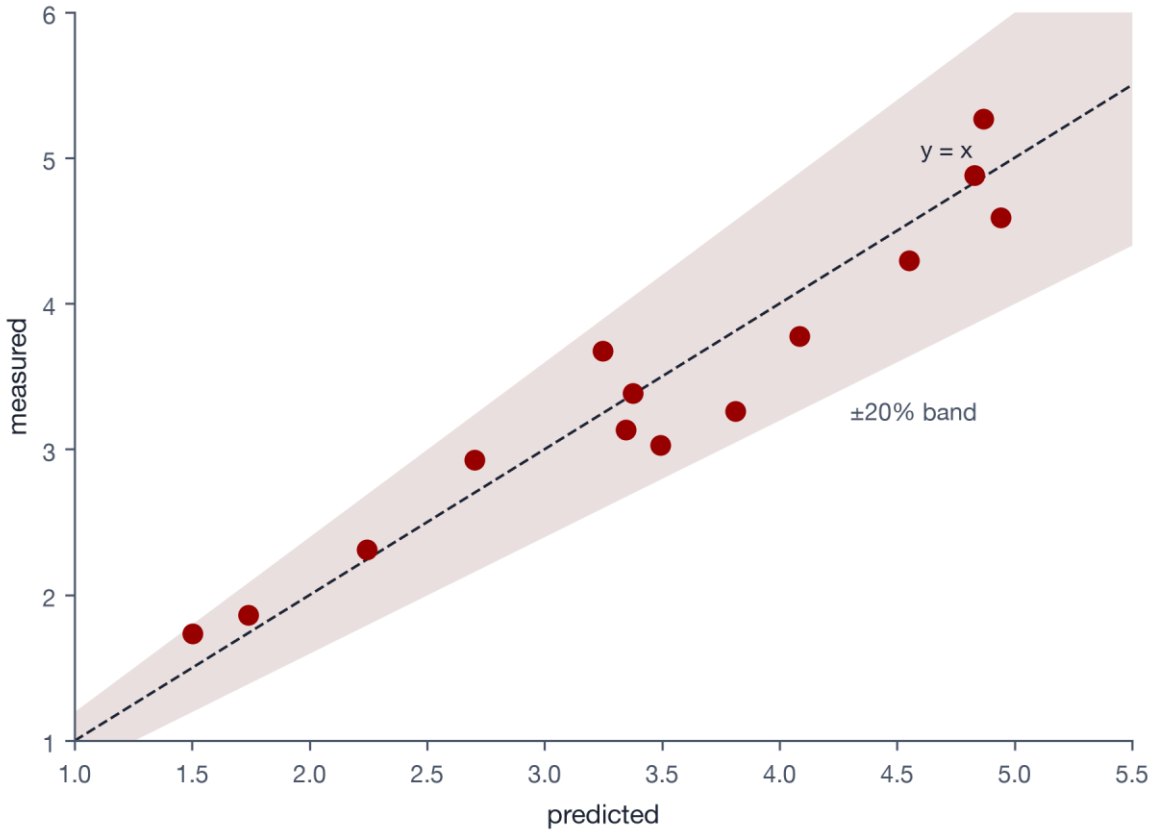
Candidate cause	Evidence (case A vs case B)	Verdict
Explanation one	Same value in both cases (0.61 vs 0.60)	Ruled out
Explanation two	Measured directly; no difference within the noise floor of the setup	Ruled out
Explanation three	Differs by 4×	Supported
Explanation four	Not measured yet; needs a new run	Open
Explanation five	Matched by design	Ruled out
Explanation six	Partial evidence only (n = 1 pair)	Weak
Explanation seven	Rejected by a control	Ruled out

Model check: the prediction tracks the measurement

First paragraph: what the graph shows and how to read its axes. Name the series and the reference line, and say which direction is better so nobody has to ask during the talk.

Second paragraph: the one pattern to notice, with the number that shows it.

Third paragraph: the limit of the claim. State the sample size, what was not tested, and what result would change the conclusion. Keep it to three lines.

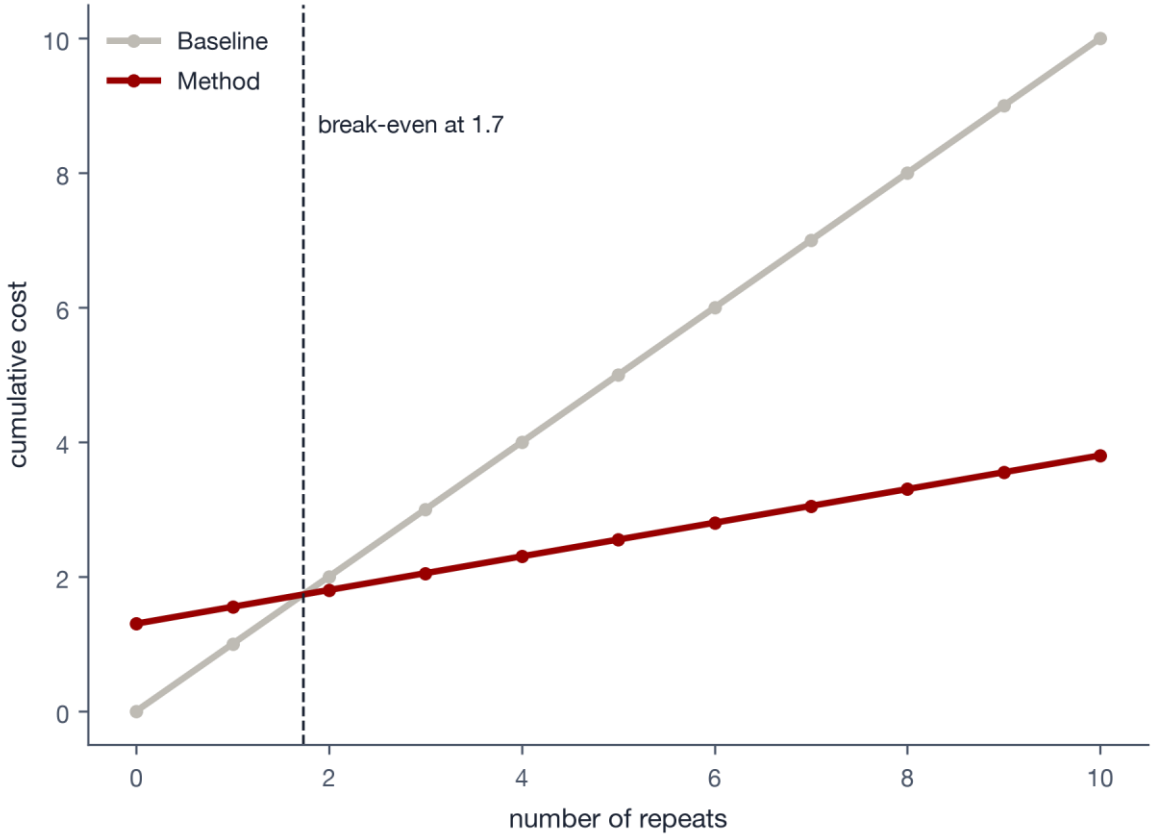


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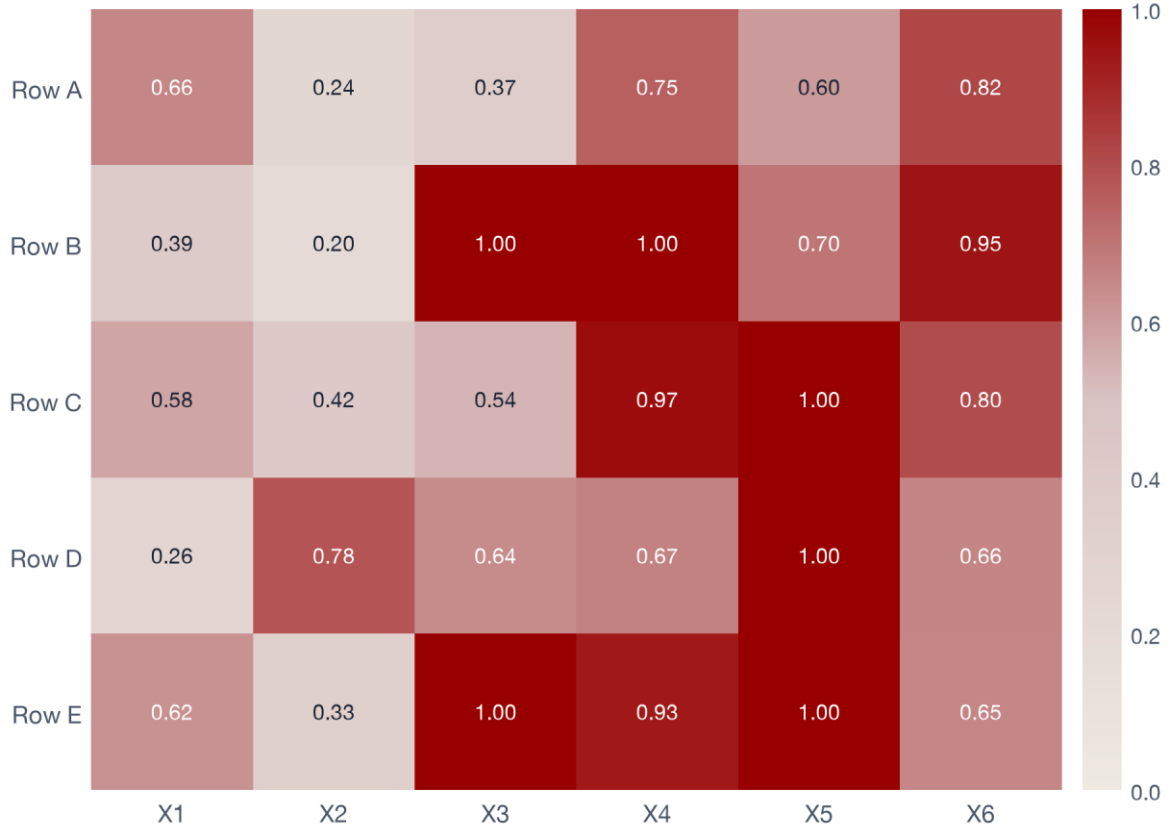


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Third paragraph: the limit of the claim. State the sample size, what was not tested, and what result would change the conclusion. Keep it to three lines.



02 · APPROACH · THREE OPTIONS + TABLE

Decision: three options, one chosen

Option A · chosen

One sentence on what this option does and why it was picked

Option B · parked

One sentence on why this option waits

Option C · later

One sentence on when this option returns

Screen used to choose (n = 6 runs)	Result
Criterion one	Option A ahead on 5 of 6
Criterion two (pre-registered)	Option A ahead; B ties; C not measured on this screen
Cost	A is cheapest; C needs new hardware

Locked choice: name the option, the metric it will be judged on, and the date it was locked.

One sentence that states the idea of this section in plain words.

Two sentences that explain it: what it assumes, what follows if it is true, and what result would show it is false.

One sentence that states the idea of this section in plain words.

Two sentences that explain it: what it assumes, what follows if it is true, and what result would show it is false.

03 · SETUP · NUMBERED STEPS + SIDE CARD

Mechanism: how it works in four steps

1. First step: what goes in and what the system does with it.
2. Second step: what is computed or compared next.
3. Third step: the key move, with the condition it depends on to work.
4. Fourth step: what comes out, and what is checked before it is used.

So: the consequence of the four steps in one line.

It accounts for

- First observation this mechanism explains here
- Second observation it explains
- Third observation it explains here
- Fourth observation it explains, if any
- What it does not explain, stated honestly here

Positioning: what prior work covers, what is new

Already covered by prior work

- Prior approach one (cite a paper)
- Prior approach two (cite)
- Prior approach three (cite)
- Prior approach four (cite here)

What survived the search: the gap that remains, in one or two lines, with the closest work named.

Planned contributions

C1 Name. One sentence on the first contribution.

C2 Name. One sentence on the second contribution.

C3 Name. One sentence on the third contribution.

03 · SETUP · FOUR CARDS

Process: four guards on the result

Pre-registration

Plans and pass rules committed before the run. Changes go in dated addenda, never edits.

Controls

A positive control that must pass and a negative control that must fail, checked in a pilot.

Provenance

Every number comes from a committed result file, with the code version and the seed.

Caught early

Name one or two problems this process caught before they reached a result or a claim.

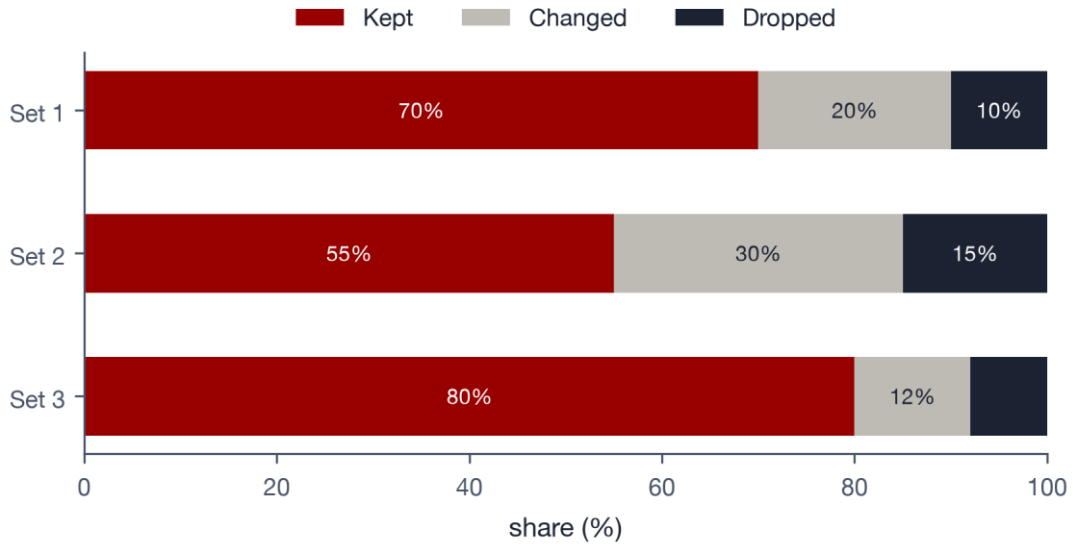
Rounds so far: what each one tested

Round	Question	Result
R1	Does the effect appear at all on the pilot?	Yes, on 4 of 5 conditions
R2	Does it hold on fresh data?	Yes, within the pre-set margin
R3	Does it survive the negative control?	Control refused as required
R4	Does it hold at full scale on the server?	Pending; job queued

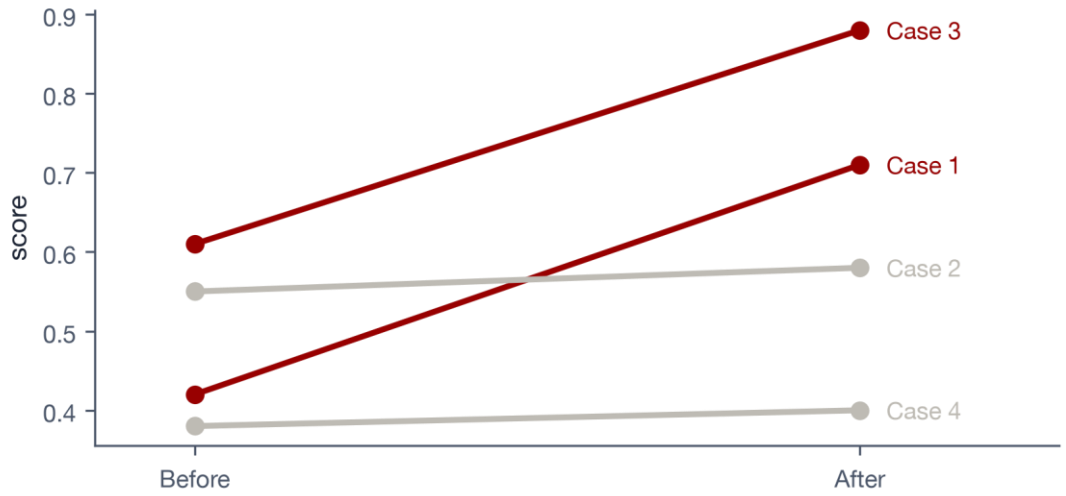
Note one: why one round failed or changed the plan, in one or two lines.

Note two: what triggered the next check or audit.

Two results side by side, one per graph

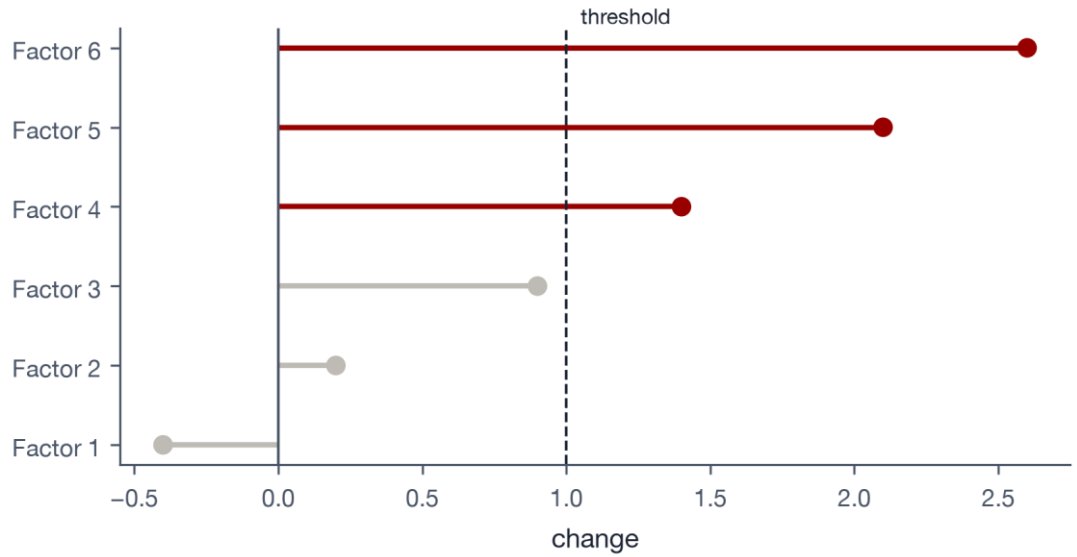


Left graph: what it shows, with the one number to remember.

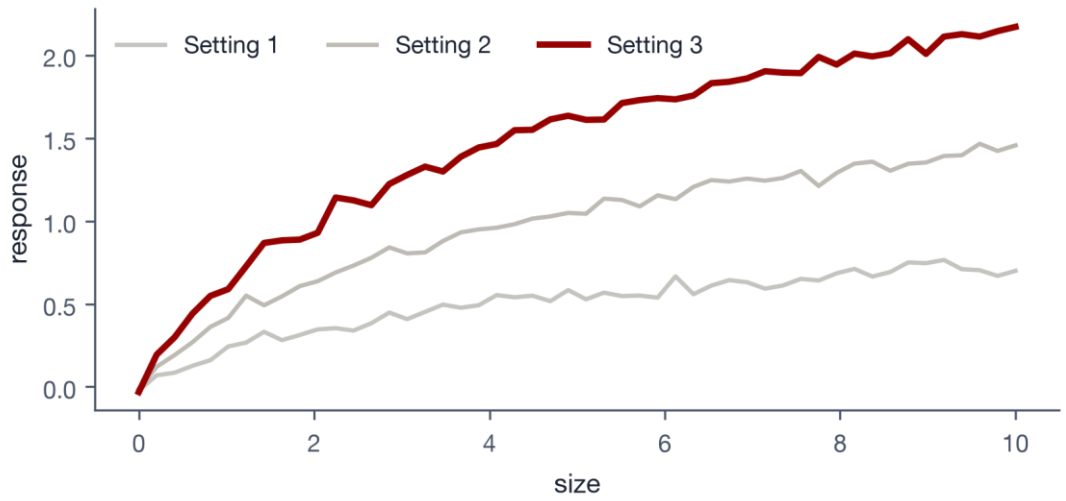


Right graph: what it shows, with the one number to remember.

Two results side by side, one per graph

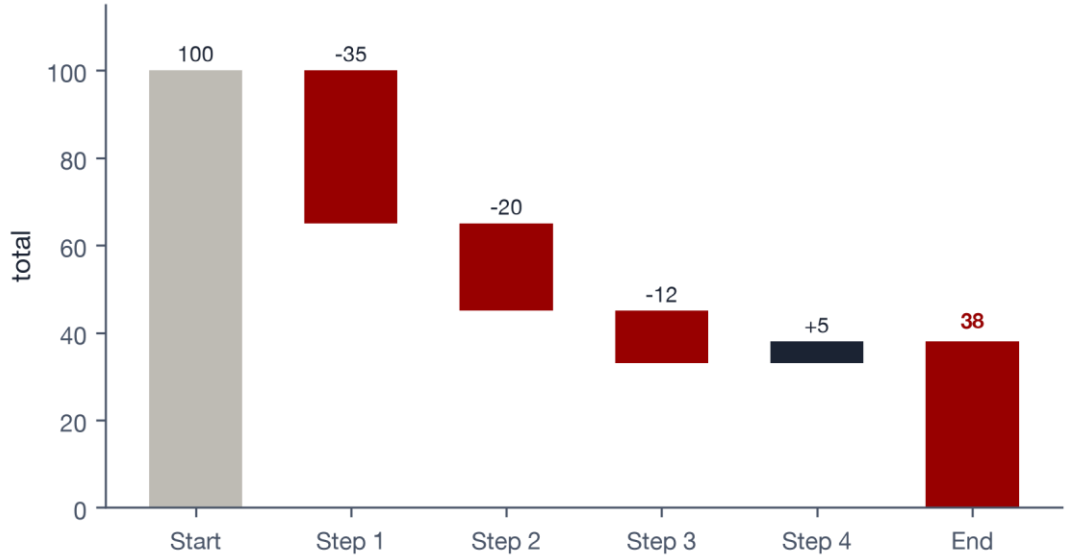


Left graph: what it shows, with the one number to remember.

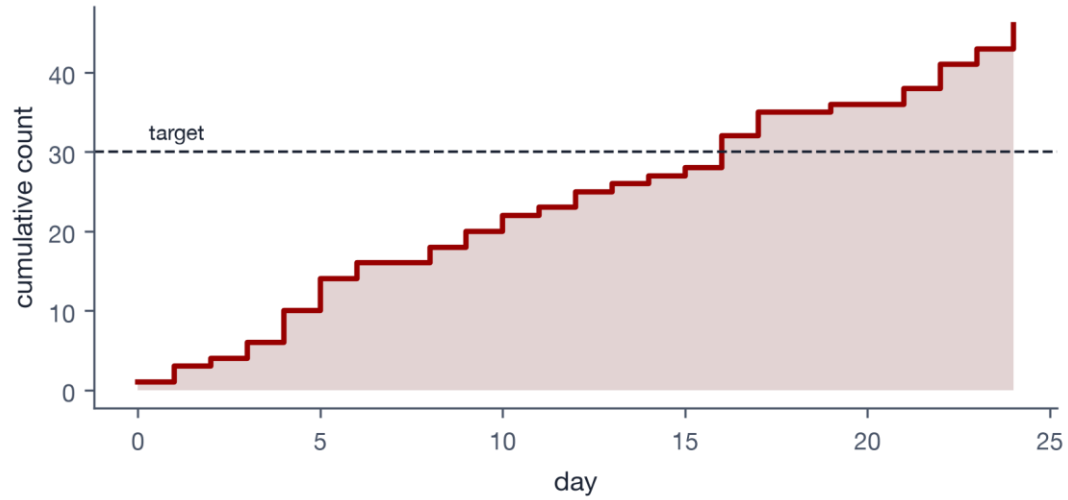


Right graph: what it shows, with the one number to remember.

Two results side by side, one per graph



Left graph: what it shows, with the one number to remember.



Right graph: what it shows, with the one number to remember.

Re-check: original vs corrected numbers

Result	Original	Re-checked	Tag
First result	0.42	0.41	Holds
Second result	0.95 [0.9, 1]	0.53 [0.4, 0.6]	Revised
Third result	12%	11% (exact)	Holds
Fourth result	0.28	0.30	Holds
Fifth result	Yes	No (artifact)	Dropped
Sixth result	4.1	4.2	Holds
Seventh result	n = 40	n = 40 + 20	Extended

Footnote: the cause of any revised or dropped row, in one line, and what now guards against it.

Three metrics, compared on one check

Metric one

0.10–0.28

What this range means and whether it passed the pre-registered bar.

Metric two

0.95–1.00

What this range means; name the condition under which it holds.

Metric three

0.94 → 0.53

What changed between the two values and why.

Lesson: one sentence the audience should carry from these three numbers.

04 · RESULTS · TWO DETAILED CARDS

Two issues found, and what each one changed

ISSUE 1 · SHORT NAME

What went wrong, in one or two lines, with the quantity involved.

How it showed up and what the corrected value is. One or two lines with the number.

ISSUE 2 · SHORT NAME

What went wrong, in one or two lines, with the quantity involved.

How it showed up, what the corrected value is, and what else the fix changed. Keep to three lines.

Bottom line: what was fixed for good and which test now guards it.

04 · RESULTS · BIG NUMBER + BULLETS

A clean negative: the effect did not appear

0 / 36

cells met the pre-registered bar across the full sweep

- First supporting point: what the measurements show instead, with a range or a value
- Second supporting point: the one exception and its size
- Third supporting point: which control rules out a broken test here
- Fourth point: what was not tested

Decision: what this result closes, and where the claim now has to rest.

Problems found, each now guarded by a test

Problem	What it faked	Guard
Problem one (name)	A speedup that was cache	Cold-cache run before every timing
Problem two (name)	Exactness on empty output	Minimum row count per sample
Problem three (name)	A pass on the wrong column	Known-bad case must fail
Problem four (name)	A stable estimate	Stock-vs-stock identity check

Box one: a test suite, its size, and its status

Box two: a tool or runner, and its effect

Box three: a registry or linter that enforces the rules

What still stands, and what doesn't

Stands

- First result that holds, with its sample size (n =)
- Second result that holds, with the evidence behind it
- Third result that holds, stated narrowly
- Fourth result that holds, with its strongest control
- Fifth result that holds, or delete this line

Withdrawn or unproven

- First claim withdrawn, with the reason
- Second claim not yet shown
- Third claim withdrawn
- Fourth claim untested, with the plan
- Fifth claim, or delete this

Compute status: what has run and what is queued

Batch	Items	Status
Batch 1	Pilot, one sample, all arms	Done, passed
Batch 2	Measured run, 9 samples	Done
Batch 3	Extra controls	Running
Batch 4	Fresh data	Queued
Batch 5	Second machine type, timing only	Queued
Batch 6	Re-run of one	Blocked, see the note

Operations note: one problem in the infrastructure and the fix now applied to every job.

Blocker: what is waiting, on whom, and what happens next.

05 · ANALYSIS · PROPOSED NEXT STEP

Name of the proposal goes here

Two or three sentences on the proposal: what it changes, why the results so far point to it, and what it would take to test. Say plainly whether it has been tested yet.

05 · ANALYSIS · TWO-COLUMN FRAMEWORK

Framework: detect first, then correct

DETECT · WHAT IS MEASURED

One or two lines on what the detector looks for and what data it uses to find it.

The quantity it reports, in one line.

What it does not need, and the control it is compared against.

CORRECT · WHAT IS CHANGED

One or two lines on what the correction changes in the system.

When it applies, in one line.

What it leaves alone, and why that is the point.

Roadmap: the next slides take each piece in turn, in the same order as the columns above.

Definition: the set the method works on

Key definition or formula goes here, on one line: $\mathcal{G} = \{ h : \text{condition, constraint} \}$

Plain-words reading of the formula, with every symbol named once, on one line.

First consequence

One sentence on what follows from the definition.

Second consequence

One sentence on what the data can't decide.

Why it matters

One sentence on why the method needs it.

Lemma: why the effect happens

Lemma. If the assumption holds, then the consequence follows for every item in the set, so the effect appears. State it in two lines.

Intuition: one line on why this is true in practice.

Key equation = Σ_i terms that make it work

What the equation scales with, and what it does not scale with.

Known

What prior work already established, with citations (Author et al. year; Author et al. year).

Claimed as new

The part of the lemma that is new, stated narrowly enough to defend in review.

Procedure: how the quantity is computed

1. **Inputs: the data and models the procedure reads, all of which exist before it runs, listed here**
2. **Step one: set the main parameter from the data, and say which data, so it is not tuned on the answer**
3. **Step two: run the search or fit, with the budget and the stopping rule stated up front**
4. **Step three: compute the reported quantity and its interval**
5. **Control: the same procedure on a random input of equal size**

ADVERSARY OBJECTIVE

max objective term

− penalty one

− penalty two

One or two lines on what the objective aims at and why each penalty is there.

Why we expect it to work on the hard case

Observation	What it implies
First observation from the earlier slides	What it implies for the method, in one or two lines here
Second observation (n =)	What it implies for the method, in one line
Third observation, short	What it implies for the method, in one line here
Fourth observation from the results	What it implies for the method, in one line, briefly
Fifth observation, short	What it implies for the method, in one line or two

Caveat: the main reason this reasoning could be wrong, in one or two lines, and the test that would show it.

The correction, as an update rule

CONSTRAINED OBJECTIVE

$$\max \text{ objective term}$$

$$\text{s.t. constraint} = 0$$

$$\lambda \leftarrow \lambda + \eta \cdot (\text{constraint value})$$

One or two lines on why the constraint is set this way and what it leaves free.

1. Step one of the implementation, here
2. Step two of the implementation, here
3. Step three of the update
4. Step four, if needed
5. Cost: extra compute per step, measured

Variant: an alternative form of the same correction, in one or two lines, and when to prefer it over the main form.

05 · ANALYSIS · COMPARISON TABLE

Why this choice, not the usual one

Question	Usual approach (cite)	This approach
What is assumed?	Worst case over all	Only what data fixes
What is changed?	Every update	Unidentified part only
What it costs	Slower on easy cases	Same speed on easy cases
What can fail?	Too cautious	Misses a case it can't see
Tested where?	Prior work	This talk, slide __
Verdict	Strong baseline; we compare	Our choice

Bottom line: the one difference that decides between the two approaches, stated in one line.

What the method should guarantee

PROPOSITION (STATE THE ASSUMPTIONS)

$$\Delta \text{ outcome(method)} = \Delta \text{ identified part} + O(\varepsilon)$$

Plain-words reading: what the guarantee rules out, in one line, with no symbols needed to follow it.

It does promise

What the guarantee covers, in one sentence.

It doesn't promise

What it leaves open, in one sentence.

Still open

What remains to prove, in one line.

Why this should not fail the way the last one did

Earlier failure	How the new design avoids it
Failure one, in a few words	The change in the design that removes it
Failure two, in a few words	The change that removes it, in one line
Failure three, in a few words	The change that removes it, in one line
Failure four, short	The change that removes it, in one line
Failure five, in a few words	The change that removes it, in one line here
Failure six, in a few words	The change that removes it, in one line

Bottom line: this removes the known failure modes; it does not remove the chance the method just fails.

How the next experiment is set up

Arm	What it rules out
Baseline	Nothing
Method (full version)	The main hypothesis
Method, no part 1	Part 1 is not needed
Method, no part 2	Part 2 is not needed here
Random control	Any change

Setting: data, task and condition, with ≥ 10 fresh seeds per arm

Primary: the metric the claim is judged on, fixed in advance

Guard: the cost on cases where the baseline already does well

After that: the next setting to test if this one passes.

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Setting: data, task and condition, with ≥ 10 fresh seeds per arm

Primary: the metric the claim is judged on, fixed in advance

Guard: the cost on cases where the baseline already does well

After that: the next setting to test if this one passes.

Where it sits, and what may be new

Prior work	Overlap	Difference
Paper one (Author 2024)	Shares the core assumption	Different goal and a different test
Paper two	Same data, same baseline	We add exactness; they approximate
Paper three (2023)	Similar mechanism	Different domain and scale
Tool or system four	Same caching idea	Ours verifies every output line
Paper five	Same metric	Measured on new hardware

Plausibly new: the one or two pieces no listed work does, stated narrowly enough to defend in review.

Tests in order, cheapest first

Test	Drop the idea if	Odds
Test 1: re-use old runs	The effect is absent on existing data	60%
Test 2: small new run	The control fails or the effect flips	50%
Test 3: full measured run	The effect is below the pre-set margin	40%
Test 4: second setting	It holds in one setting only	30% (ok)

Cost note: what the cheapest test needs and how long it takes to run.

Risk: the one thing that could void the whole plan, and how it would show up.

06 · NEXT STEPS · NEXT TWO WEEKS

Next steps and asks

Next

1. First task, with an owner (date)
2. Second task, with a date
3. Third task, which depends on the first
4. Fourth task, if time allows
5. Fifth task, or delete this line

Asks

- First question for the audience, stated so it can be answered
- Second question or request
- Third question or decision
- Fourth ask, or delete this line